

2026-08-17 Update: Reader Prompt Fix (Comparison-Question Answer Misalignment)

1. Finding the Problem: Reading Through Concrete Cases, Not Just Aggregate Metrics

After the gate three-signal ablation (0810 update) was completed, instead of continuing to tune aggregate metrics, a case study was run on arm D (feature-level fusion, MuSiQue dev full set). Wrong-answer cases were first split by whether retrieval was fully correct:

	Count
Wrong answer (EM=0)	1,376
— of which: every hop retrieved the gold passage, but the final answer was still wrong	485

These 485 cases, where retrieval and the gate both did their job correctly and the answer was still wrong, have nothing to do with retrieval or the gate — the problem is in the last step, where the reader synthesizes all intermediate results into a final answer. Comparing each of these 485 final answers against the per-hop short answers (prior) yields three categories:

Type	Count	Share
Final answer equals the last hop's short answer, but still graded EM=0 (pure wording mismatch against gold, e.g. "McKinley" vs. "President McKinley")	372	77%
Final answer equals an earlier hop's short answer (the reader genuinely picked the wrong hop's conclusion)	39	8%
Matches neither (the reader synthesized a new, wrong answer on its own)	74	15%

The first category is an artifact of EM's strictness, not a real error. The second and third are the real problems, together about 23%.

Repeating the same analysis on 2WikiMultihopQA and HotpotQA showed the "reader picked the wrong hop" pattern is **not MuSiQue-specific — its share is even higher on the other two datasets:**

Dataset	Retrieval fully correct, answer wrong	— of which "reader picked wrong hop"	Share
MuSiQue	485	39	8.0%
2WikiMultihopQA	4,894	639	13.1%
HotpotQA	1,522	232	15.2%

Digging further into the 2WikiMultihopQA and HotpotQA cases showed "picked the wrong hop" concentrates heavily on **comparison-style questions** (2WikiMultihopQA has a dedicated comparison question type) — and it isn't simply "picked the wrong hop," the reader isn't performing the comparison at all, and is directly outputting a raw fact from some hop (usually a date) as the answer:

Q: Which film was released more recently, Royal Treasure or When Love Begins?
gold: Royal Treasure | final: January 15, 2016
per-hop short answers (prior): ['January 15, 2016', '2008']

Q: Did the board game San Marco or About Time come out first?
gold: San Marco | final: 2001
per-hop short answers (prior): ['2001', '2007']

The two hops correctly looked up each film's/game's release date, but the answer that should have been output is "which one" (the title/name) — the reader just echoed one of the raw dates verbatim instead.

2. Root Cause and Fix: The Reader Prompt Was Missing an Explicit Comparison-Question Instruction

The final-answer generation prompt is built by

`retrieval/run_retrieval_exp_wavefront.py::build_final_reader_cot_prompt`. Its only sentence constraining answer format is:

Give a short span or phrase only (entity, date, number, or yes/no).

This explicitly lists "date" and "number" as **valid answer types**, treated identically across all question types. For most bridge questions, the assumption "the last hop's short answer IS the final answer" holds; but for comparison questions, the dates/numbers looked up at each hop are only **intermediate values used to make the comparison** — the actual answer should be the name of whichever entity wins the comparison. The prompt never distinguished these two cases, so the reader frequently just output a hop's raw date as the final answer.

The fix adds a new function to `run_retrieval_exp_wavefront.py`,

`build_final_reader_cot_prompt_comparison_hint` (**the original function is left untouched**, the new one sits alongside it), appending exactly one sentence to the base prompt:

If the original question asks you to COMPARE two or more entities (for example "which was released more recently", "who was born first", "which is longer"), the per-step answers above are the values used to make that comparison, not the final answer themselves -- you must perform the comparison yourself and answer with the NAME of the entity that satisfies it, not a date, number, or other raw value.

`retrieval/run_retrieval_exp_wavefront_gate_v3_rawprefix.py` gained a new `--final-reader-prompt {default,comparison_hint}` flag to switch between the two versions, with everything else (retrieval, gate scoring, per-hop candidate selection) left completely unchanged — guaranteeing a clean single-variable comparison.

3. Results: 2WikiMultihopQA Full-Set Validation

MuSiQue dev, $\lambda=0.50$, rawprefix, with only `--final-reader-prompt` switched:

	recall@1	chain@1	EM	F1
default prompt	0.9289	0.8613	0.4990	0.5759
comparison_hint	0.9289	0.8613	0.5982	0.6770
Δ	flat	flat	+9.92	+10.11

recall@1 and chain@1 are completely unchanged — retrieval and the gate weren't touched at all, proving the gain comes cleanly from this one prompt change. EM/F1 rose by roughly 10 percentage points, the single largest gain from any change in this project so far, larger than λ tuning, rawprefix text alignment, and the BGE retrieval prefix combined.

This also confirmed the root-cause diagnosis: the "reader picked wrong hop" bug's reproduction rate dropped from 13.1% (639/4,894) to **2.6%** (98/3,833), a 5x reduction.

4. MuSiQue and HotpotQA Validation — Completed

All three datasets are now validated. recall@1/chain@1 stay exactly flat against the default prompt on every dataset (retrieval and the gate are untouched, confirming the change is clean again), and EM/F1 are positive across the board:

Dataset	recall@1	chain@1	EM (default→comparison_hint)	F1 (default→comparison_hint)
MuSiQue	0.7018 (flat)	0.5143 (flat)	0.4307→0.4427 (+1.20)	0.5270→0.5386 (+1.16)
2WikiMultihopQA	0.9289 (flat)	0.8613 (flat)	0.4990→0.5982 (+9.92)	0.5759→0.6770 (+10.11)
HotpotQA	0.6596→0.6595 (flat)	0.5568→0.5567 (flat)	0.5217→0.5517 (+3.00)	0.6517→0.6870 (+3.53)

The size of the gain ranks 2wiki > hotpot > musique, which doesn't exactly match the ranking of the "picked wrong hop" bug's **share within affected cases** (hotpot's 15.2% is higher than 2wiki's 13.1%) — the reason is that hotpot's retrieval baseline is lower to begin with (recall@1 only ~66%), so the "retrieval fully correct" precondition covers fewer examples; what actually drives the overall EM gain is the number of affected cases **as a share of the full dataset**, and there 2wiki (639/12,576 = 5.1%) is higher than hotpot (232/7,405 = 3.1%).

All three datasets moved in the same, positive direction with no regressions anywhere, so `comparison_hint` was adopted as the new default prompt.

5. Further Exploration — Tried, Not Adopted; `comparison_hint` Is Final

Continuing the case study on top of `comparison_hint` (the residual "retrieval fully correct, answer still wrong" cases on 2WikiMultihopQA) surfaced two categories of problems `comparison_hint` still didn't cover, and a new prompt fix was tried for each. Both were validated on all three datasets using `replay_final_reader.py` (reconstructs each hop's input from the `prior/para_ids` already stored in an existing case file and only reruns the final-answer step — retrieval, the gate, and the per-hop short answers are never rerun, which is much cheaper than a full end-to-end run). Both were net-negative and were not adopted:

- `prompt_short_answer_with_context_type_match` (per-hop short answer, constrains the answer's type to match the interrogative word): helped on MuSiQue (EM +1.94/F1 +2.32, on top of `comparison_hint`), but was roughly flat on 2wiki with a small drop on the retrieval side, and roughly flat on HotpotQA — inconsistent across the three datasets, not adopted.
- `build_final_reader_cot_prompt_comparison_reasoning` (require an explicit reasoning line before the final answer): helped on 2wiki (EM +2.65/F1 +2.31), but was net-negative on both MuSiQue (EM -1.32/F1 -1.29) and HotpotQA (EM -2.35/F1 -2.92). The case study found the root cause: the model was treating `comparison_hint`'s "answer with the entity's name, not a date/number" instruction as a rule to apply unconditionally, invoking it whenever a name and a date both happened to appear in its own reasoning — even for questions that weren't comparison questions at all (a striking example: the model correctly derived "June 1982" as the answer, then overrode it to the person's name "Diego Maradona" purely because of this rule).
- `build_final_reader_cot_prompt_comparison_positive` (rewrote the same rule in positive form and dropped the illustrative examples): net-negative on all three datasets, with the largest drop on 2wiki (EM -3.05/F1 -3.12). The case study found the opposite root cause this time: without the illustrative examples, the model failed to recognize many comparison questions as comparison questions at all, so the trigger rate collapsed and the reader reverted to the original `comparison_hint`-era problem (echoing a raw date from one hop instead of comparing).

Conclusion: `comparison_hint`'s existing wording (examples included, "not a date/number" clause included) already sits at a reasonably good balance between **triggering on real comparison questions** and **not over-triggering on non-comparison ones** — each of the two follow-up attempts changed only one half of that balance and made things worse. Further local patching of this prompt has diminishing returns; `comparison_hint` is the final version, and this line of exploration is closed.

Next steps: backfill every number in `gate_v3_main_method_report.md` that references the "current complete method," and consider rerunning arm B (gate v2 rawprefix) and arm C (h-only rawprefix) with this new prompt as well.